Introduction

Biotic stresses like insect pests, diseases and weeds cause serious yield losses in rice. Nearly 33% of total rice yield is lost annually due to pests alone. About 20-40% of the world's potential crop production is lost annually because of the effects of weeds, pests and diseases[1]. Deterioration of rice grain/seed quality due to several bio-agents in the paddy fields as well as during storage poses a threat to the very existence of large human population, as it directly affects rice yield considerably in addition to the other biotic stresses. New pest problems continue to develop. Attempts to control agricultural pests have been dominated by chemical control strategies, but indiscriminate use of agrochemicals has led to problems of human health, erosion of soil quality, environmental pollution and degradation of natural resource base. Rapid growth of agriculture is essential not only to achieve self-reliance at national level but also for household food security and rapid reduction in poverty levels. National Policy of Indian agriculture seeks to promote technically sound, economically viable, environmentally non-degrading and socially acceptable methods of pest management for sustainable agriculture. Intensive use of land since 1960s for enhanced crop production through the use of agrochemicals was mandatory to attain food security in India. Paradoxically, the ancient wisdom that low cost techniques have sustained food production for several millennia was ignored in our approach. Demands for ensuring food forever rising population growth continue to drive the land use for still greater intensive food production. Ancient wisdom based on certain key factors such as living in harmony with nature, has the answer to attain sustainable agriculture[2]. This is the right time to rediscover our past heritage of sustainable agriculture and blend it with the current agriculture.

Pesticide use is capital intensive as it involves purchase of the pesticide, sprayer and personal protective equipments (PPE). It is difficult for farmwomen to use above expensive inputs because generally they get small returns from marginal farming. Women's limited access to productive resources often makes them more reluctant than men to purchase inputs such as pesticides for use on their crops, which are usually food crops. Use of capital intensive technologies are least beneficial for the poor, including women and children. Recent research in India shows that small scale and marginal farmers take loans from private finance corporations to purchase inputs and then, unable to pay their debts, become answerable to moneylenders, moreover marginal farmers have a 10 times greater risk of severe pesticide poisoning then large scale farmers[1]. Women in small and medium scale farming suffer the worst health problems from pesticide use because they spray the fields themselves, usually without safety precautions.The rapid physiological changes experienced by women during pregnancy, lactation and menopause render them more vulnerable to toxins. Exposure to pesticides can cause miscarriage, pre mature birth, birth defects and low birth weight. A substantial portion (up to 33%) of a woman's chemical burden can be passed on to an unborn child during gestation and to a baby through breastfeeding. Use of such pesticides is prohibited or severely restricted in OECD (Organization of Economic Cooperation and Development) countries, and in line with the International Code of Conduct.

In India, with the introduction of photo insensitive and fertilizer responsive high yielding rice varieties (HYV) after the initiation of green revolution in mid sixties, the insect pest scenario became more complex due to altered microclimate[3]. Regular pests like stem borer and gall midge have become serious problems in many new areas. Overall, yield loss due to insect pests is estimated to range from 21 to 50% mainly due to stem borer, gall midge, leaf folder and brown plant hopper. Yellow stem borer (YSB) is widely distributed and causing serious damage as high as 80% in late planted crop particularly in Punjab, Haryana, Uttar Pradesh, Orissa, Andhra Pradesh, Maharashtra, Tamil Nadu, Kerala and Karnataka.

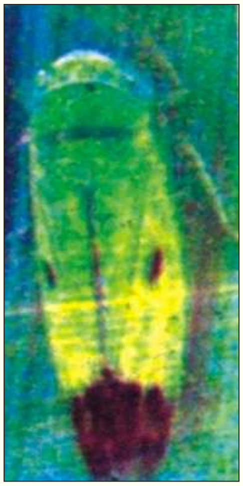
Women are the major decision-makers in ensuring nutrition to the next generation as they provide primary nutrition to the young children. Women play an important role in Indian agriculture since Stone Age as man was collecting their food by hunting up to 20% while women managed rest 80% through root, fruit and tubers of the plant. Women farmers are utilizing their maximum time up to 15 hours for agriculture or agriculture related work therefore, they are being considered as backbone of Indian agriculture. Women contributed in almost all the activities right from nursery raising to the post-harvest activities in rice. In some activities like drying of produce, winnowing and cleaning of grains, the women participation gone over 75%. Transplanting and harvesting, however, got the maximum average hours i.e. 43.13 and 26.13 /women/season, respectively[4]. Crop protection strategies - the management of pests, diseases and weeds have changed dramatically over time. The intensification of agriculture alters agricultural practices significantly. In intensive agricultural systems, more traditional and labour intensive physical and biological crop protection measures are superseded by more capital intensive use of pesticides. Pesticides can increase agricultural productivity, but when used injudiciously, they are toxic to humans and other species. A study conducted by FAO, WHO and UNEP broadly estimates that between 1 million to 5 million cases of pesticide poisoning occur each year, resulting in several thousands fatalities. Pesticide fatalities are overwhelmingly a developing country phenomenon. Although developing countries use just 25% of all pesticides produced, 99% of deaths from pesticide poisoning occur in developing counties. Children and women are specially at risk. So, there is an urgent need to empower women in gender friendly technologies of pest management. This will be a vital input for planned growth and sustainable development of agriculture in the country. The present document discloses low cost, gender friendly, indigenous techniques, which is beneficial for the sustainable pest and disease management of rice.

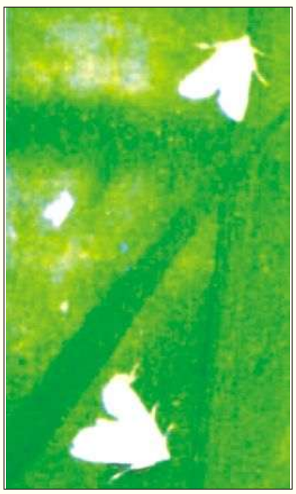
Low cost pest management techniques in rice

Pest	Technique/ practice	Region
Asian rice gall midge *Orseolia oryzae* Wood Mason  Gall midge maggot	Farmers collect and spread *pasu* (*Cleistanthus collinus*) leaves @ 10kg / 100m^2 area in the infested field for controlling gall midge. Economic threshold level (ETL) of this pest is 1 silver shoot / m^2 or 5% affected tillers in the field.	Tamar block of Ranchi district in Jharkhand
	Spreading of fresh leaves of *salai* (*Boswellia serrata*) @ 5 kg / 100 m^2 in the infested field is also reported to control gall midge[5].	Himachal Pradesh
	Farmers adopt very early planting with short duration varieties for last few years, which resulted in low gall midge activity in ARGM endemic areas. Judicious use of nitrogen especially during tillering phase and also alternate wetting and drying the fields minimize incidence of gall midge[6].	Sambalpur area of Orissa
	Seed treatment with chlorpyrifos or isofenphos (0.2% solution) for 3 hours or seed mixing with chlorpyrifos (0.75kg a.i/100 kg seeds) provides protection for 30 days in the nursery and subsequently reduces gall midge incidence in the transplanted crop or direct seeded rice up to the vulnerable stage of the crop[6].	
	Seedling root dip has proved most effective and economical. Chlorpyrifos, isofenphos or chlorfenvinphos at 0.02 to 0.04% concentration for 12 hours seedlings root dipping offered protection up to 30 days after transplanting[6].	
 Gall midge adult	Farmwomen can grow resistant varieties like Shakti, Surekha, Phalguna, Kakatiya in endemic areas of the gall midge infestation.	

Rice field crab *Paratelphusa hydrodromus* H. 	After first rain, she crab brings along young ones and leaves them in shallow waters of paddy fields. Crabs damage the crop more in the initial days of the monsoon after transplantation. They nip off stem near the surface of the ground after 30-35 days of transplanting. Rainy season (*Kharif*) paddy crop is prone to attack by the crabs. Crabs also burrow the bund leading to percolation of water.	
	Farmers cover and block the crab holes in bunds by the powder of *Butea monosperma* flower commonly known as *flame of forest*, *palsh, khakra, modnga* and *parasa*. Results are visible with in 12-24 hours [7 and 8].	Gujarat and Karnataka
	Farmers soak seeds of *Tamarinda indica* commonly known in Indian language as *imli, tentul, chinch, amli, chinta- chette, puli, huli, tentuli* and *imbli*, in water for 24 hours and broadcast on bunds and hedges of the field to control crab menace. Seed get firmly lodge in crab's mouth and they get chocked and ultimately die within a day or two[9].	Valsad area of Gujarat
	Poison baits with warfarin @ 0.025% in puffed rice mixed with fried onions and fish can be kept @ 3g/hole on the bund for three weeks for effective control of rice field crab[10].	Orissa
	Application of *Asafoetida* (*Ferula asafoetida*) commonly known as *hing, ingu, inguva, perunga & hengu* @ 1 g / 10 m^2 at 45 days of planting in paddy fields controls crabs by emitting strong odour[8].	Karnataka
	In order to control crabs in paddy fields, raw cow dung @ 300 kg /ha is used in standing water in lowland areas. This disturbs movement of crabs and also produces unbearable odour to them, which causes crabs to go out of the field[11].	Uttar Pradesh and Orissa

Brown plant hopper *Nilaparvata lugens* Stall. 	Brown plant hopper is a serious and destructive pest causing immense damage resulting in loss of yield and grain quality. Close planting, production of more tillers per unit area, increased use of nitrogenous fertilizers and indiscriminate use of pesticides are reported to increase the abundance[12] . Economic threshold level (ETL) of this pest is 10 hopper/ hill.	
	Control of brown plant hopper in rice is possible by dusting of ash on the standing crop[11].	Uttar Pradesh
	Leaves of *Calotropis gigantea* are pressed and incorporated into the soil in the interspaces available. It controls brown plant hopper in nursery as well as in the field.	Pudupattim, Tamil Nadu
	At nighttime, two torch lights are beamed in a 'V' shape in the center of the paddy field. The person holding the torches walks from the center to the edge of the field. The hoppers are attracted to the light and attempt to follow it. Thus, they leave the paddy field. This process is repeated for two or three days in succession[13].	Karnataka
White backed plant hopper *Sogatella furcifera* Horvath.	Spray of *Metarhizium anisopliae* @ 10^7 spores/ml and *Beauveria brongniartii* @ 10^5 spores/ml reported to be effective in order to control brown plant hopper[14].	Orissa
	Leaf extract of *Lasiosiphon eriocephalus* commonly known as *mukudda, mukute* is very effective in controlling the brown planthopper menace in paddy. One kg of leaves is boiled in 10 litre of water, filtered and diluted to a ratio of 1:10 and then sprayed on crop, once during nursery stage and again after transplantation[15].	Karnataka

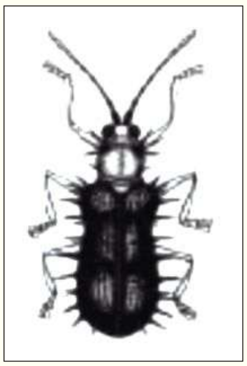
Green leaf hopper *Nephotettix virescens* Dist. 	Tribals use *water pepper* (*Polygonum hydropiper*) @10g / liter (5kg / ha) mixed with 2 ml liquid soap per litre, for the management of brown plant hopper[16]. They named it as *Gotkinamaru* (cattle-tick-killer), *Kalatadi or Galpudi*. Leaves of *Calotropis gigantea* are pressed and incorporated into the soil and found effective to control brown plant hopper in nursery as well as in the field. Economic threshold level (ETL) of green leafhopper is 1-2 hopper/ m^2 or 20 insect / hill.	Orissa
Grasshopper *Hieroglyphus* spp. 	Grasshopper is brushed with leaves containing branches of *Boswellia serrata* commonly known as *Indian olibanum, salai, parangi, sambari, madi* etc. Branches are also placed in field at the distance of 6-8 m. This is done in the evening when some water is available in the field. Success up to 70 - 80% has been reported[17].	Uttar Pradesh
	Erection of one meter high stand in the paddy fields during night and hanging of one lamp from the stand after covering it with a funnel shaped iron sieve is found effective to control grasshopper. Grasshoppers are attracted by the light and get collect around the stand. Early in the morning farmers collect the insects and dig them in the soil.	Eastern Uttar Pradesh
 Grasshopper found in paddy field	To protect paddy from viruses and grasshoppers farmers grow *Sesbania aegyptiaca* commonly known as *jayanti* as a hedge crop all round the field[18].	Uttar Pradesh

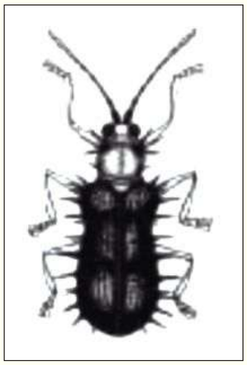
Rice case worm *Nymphula depunctalis* Guence  Case worm larva inside the leaf	Broadcasting of sand and kerosene or spreading *Bauhinia variegata* commonly known as *Bauhinia* and *Kachnar* controls paddy case worm[19]. Economic threshold level (ETL) of case worm is 2 fresh affected leaf / hill.	Jharkhand and Uttar Pradesh
 Case worm adult	Tender bamboo (*Bambusa arundinacea*) shoots rhizomes locally known as *Karil* are immersed in water for 2-3 days @ 1 kg / 4 l of water. The extracted solution sprayed in the paddy fields is reportedly controlled the caseworm. About 50 litres of the extract is required for spraying of one acre crop. The worms are controlled within 6-7 days of application [11].	Jharkhand
	Spreading of fresh leaves of *pasu* (*Cleistanthus collinus*) and *salai* (*Boswellia serrata*) @ 5kg leaves / $100m^2$ in the insect infested field is reported to check majority of insects.	West Singhbhum of Jharkhand
 Leaf tip showing cut ends and scrap of leaf tissues damaged by Caseworm	Burning of bicycle tyre is one of the practices to attract and kill the case worm in rice. Each bicycle tyre is cut into 4 pieces, requiring 20 tyres /ha. The tyre pieces are tied in pegs above vegetation heads and are burned in the evening during the incidence of pests[20]. It is not a healthy practice as it may lead environmental pollution by release of harmful smoke.	Madhya Pradesh and Uttar Pradesh
	Chopped pieces of colocasia (*Colocasia esculenta*) and chopped peels of citrus (*Citrus grandis*) are spread in the rice fields. This makes water bittersome and acts as repellent to the caseworm larvae when they fall on the water[11].	Orissa

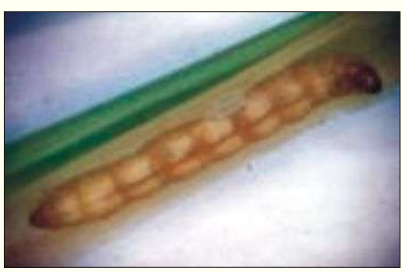
	Moving kerosene soaked rope over the crop and draining out available standing water at early tillering stage minimizes incidence of caseworm. Cases made by case worm drop down from crop on standing water and are then drained out from field.	Uttar Pradesh
Case worm larvae	Leaf folder on rice is controlled by pressing 5 - 6 bundles (10 -12 plant/ bundle) of *kalothia* (*Tephrosia purpurea*) into the mud. Plants get decomposed in the mud and release some bio-chemicals, which might act as repellent or kill the pest.	Orissa
	Leaves of *sindwar* (*Vitex negundo*) are boiled in water and cooled before spray. It is used @1kg leaves / 5 l of water for 0.06 acre, to control the caseworm. Success of this practice is reported to be 60%.	Ranchi in Jharkhand
	To control leaf folder in paddy, ducks are released in the field and they feed on the pest[11]. Economic threshold level (ETL) of leaf folder is 2 fresh affected leaf / hill.	Karruppathevan patty village in Theni district of Tamil Nadu
	Spraying of kerosene @ 5 l / ha mixed with soap and water is one of the common practice to control the leaf folder.	Madurai district of Tamil Nadu.
	Spraying of chilli (*Capsicum annumm*) and tobacco (*Nicotiana tobacum*) *extract* is reported to be effective for the control of leaf folder[5].	Nachalur village in Karur district of Tamil Nadu
Case worm adult	Branches or twigs (4-5) of fishtail palm are installed in the paddy field at the time of infestation of leaf folder. These fishtail palm branches harbour the predatory birds and control the paddy leaf folder within 8 - 10 days.	Koraput, Orissa

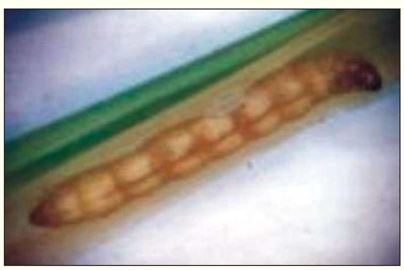
 Case worm larvae	Wild sugarcane *Saccharum spontaneum* twigs of 4 - 5 feet height and 4 - 5 cm diameter are planted after 15 days of transplanting in rice field for control of leaf folder. These erected branches harbour predatory spiders, which are active at the time of occurrence of leaf folder, there by suppressing the incidence of pest.	Benakunda village in Ganjam district of Orissa
 Case worm adult	Bt formulations e.g. halt and biolep @ 1kg/ha were equally effective (about 88% mortality) with some of the popular insecticides viz. chlorpyriphos (0.5kg a.i/ha), chlorpyriphos 50% + cypermethrin 5%, cartap (0.5 kg a.i/ha), imidachloprid (0.1 kg a.i./ha), monocrotophos (0.5 kg a.i./ha) giving about 41, 61, 75, 73 and 88% mortality of leaf folder larvae, respectively, in the field. The results convincingly proved that they could be used by the farmers for control leaf folder.Bt formulations gave more gross income than the insecticides, except for monocrotophos. Reduction of cost of the Bt formulations, therefore, would encourage bio-pesticide application for the benefit of the farmers. Aqueous suspension of both of the Bt and the fungal formulations could be sprayed @ 1 kg/ha. As most of these products are not harmful to human being, they can be used without much precaution. Bio-pesticides are self-perpetuating and therefore have long term effect[14].	Orissa

	Young bamboo (*Bambusa arundinacea*) shoots or sprouts (250g) are collected and soaked overnight in 1 litre water. In the next morning, the water is decanted and sprayed on the infected field of 1 acre. After 7 - 8 days, leaf-folder pest is controlled due to alkaline substance present in young shoots or sprouts.	Dalaiguda village in Koraput district of Orissa
	Thorny branches of *ber* (*Zizyphus* spp) are used to shake plants in the field at tillering stage. Folded leaves get opened; leaf folder gets injury and face disturbance in movement.	Uttar Pradesh
	Release of *Trichogramma japonicum* @ 50,000 - 100,000 eggs/ha is very effective technique of leaf folder management. These eggs are available in the form of *Tricho card*. Each *Tricho card* contains 20,000 eggs of parasite *Trichogramma japonicum. Tricho card* can be further divided in 10 small pieces which can be easily tagged underside of leaf with the help of alpin/locally available thorn during evening hours.	
Rice hispa *Dicladispa armigera* Oliv.  Adult hispa with white scraped symptom on leaf	White patch on leaf caused by hispa is very common in paddy crop. Farmers control hispa by dusting the sand and kerosene oil mixture (5:1). The success rate of this practice is almost 20 -25%. Economic threshold level (ETL) of rice hispa at early transplanting is considered as 1 adult or 1 damaged leaf/hill and at mid-tillering 1 adult or 1 - 2 damaged leaves per hill.	Oraina village in Nawada district of Bihar

	Farmers believe that bitter and acrid taste of the leaves and unripe fruits of *keondi, kendu, tendu* (*Madhuca latifolia*) become toxic for hispa in paddy fields. This practice is successful in controlling the pest up to 95-98%[5].	East Singhbhum district of Jharkhand
	Broadcasting the branches of *Germany ban* (*Eupatorium odoratum*), *Bihalongoni* (*Arthyrium spp*) and *Posotia* (*Vitex negundo*) on standing water at early tillering stage of rice controls hispa in rice.	Jharkhand
	Fresh leaves of parasi (*Cleistanthus collinus*) tree are broadcasted in paddy fields with standing water for the control of rice hispa. Parasi leaf has strong bitter taste, which is toxic to the insects, thus it works as repelling agent.	Jharkhand
Adult hispa	Spraying of tobacco leaf extract at mid tillering stage controls the attack of hispa. Alkaline nature of tobacco leaves is responsible for insecticidal action. Plucking of infected leaf tips in nursery also minimizes damage of hispa in the main field.	Uttar Pradesh
	For controlling rice hispa in paddy field, brooms are prepared out of dried twigs of wild *ber* (*Zizyphus spp*). The broom is stuck on the crop, at tillering stage, affected with rice hispa. As a result the grubs of hispa are injured resulting in disturbance in the movement and they fall down on the standing water. The grubs are removed from the field by draining the water. This is 60 – 70% effective and is an eco-friendly practice.	Kesinga block in Sambalpur district of Orissa
	If nursery beds are flooded, the beetles float and can be swept together with brooms and can be killed [21].	Orissa

	Prompt ploughing of fields after harvest of rice eliminates the breeding of pest in the sprouts from the stubbles and hibernating insects get destroyed[21].	Orissa
	Clipping of rice seedling tips before transplanting minimizes the hispa infestation. In this process, the eggs and grubs of hispa are discarded in nursery itself[21].	Orissa
 Adult hispa	Split application of potassic fertilizer improves the plant ability to resist the build up of rice hoppers and hispa. On the contrary overdose of nitrogen favours hispa infestation[21].	Orissa
	Seedling root dip in chlorpyriphos 20 EC @ 1ml in 1 litre water for 12 hours before planting protects the crop for 21 days in the main field[21].	Orissa
	Broadcasting of goats' excreta at mid tillering stage repels hispa and they fly away due to disagreeable odour.	Uttar Pradesh
Yellow stem borer *Scirpophaga incertulas* Walker  Yellow stem borer larva	Mass trapping by installation of pheromone trap for two month @ 20 traps/ ha is found effective in the management of stem borer. Lures used in the pheromone trap should be changed after every 15 days. Economic threshold level (ETL) of this pest is 5% dead heart or 1 egg mass / m^2 or 1 adult moth / m^2 in the field.	
 Yellow stem borer adult	Intercropping of one row of Pusa Basmati with 9 rows of main crop gave higher yields with lower white ear percent in the main crop.	Orissa

 Damaged ear head by stem borer	Spraying of neem oil @ 0.2% is practiced for the control of early stem borer in rice. Three sprayings are required after appearance of the pest, i.e. one month after transplanting The active ingredients (limonoids), particularly azadirachtin and meliatriol are found in neem and known to work as pest repellent[22].	Ganjam areas of Orissa
	Cashew (*Anacardium occidentale*) nutshell oil acts as larvicide and biopesticide. It is used for control of stemborer in rice. It is sprayed 2 - 3 times at the time of incidence[11].	Nandapore village in Koraput district of Orissa
	Yellow stem borer (*Scirpophaga incertulas*) can be controlled by seedling root dip in chlorpyriphos 20EC @ 1ml in one litre water for 12 hour before transplanting[10].	Orissa
 Rice stem borer	The farmers have devised their own means to control stemborer infestation by using *dhatura* (*Datura stramonium*) stem and leaves in the paddy field. The indigenous tribal people of the state are normally not very keen to apply chemical insecticides/pesticides, mainly due to its cost factor as well as attitude of the people to grow paddy mostly under organic farming system. The practice is to spread *dhatura* leaves and stems (cut pieces) in the paddy field particularly on observing the stem borer attack. The water in the field is then blocked through bunds so that the *dhatura* stems and leaves get decomposed. These decomposed	Dimapur and Kohima districts of Nagaland

	leaves and stems get circulated throughout the field, which acts as a repellent to the stem borer. Alternatively, the farmers drain out water from the paddy field when the infestation occurs, and it is allowed to dry up, after which fresh water is pumped in again.	
	Neem cakes are filled in gunny bags and immersed in irrigation channels for the control of stem borer [23] . The neem cake is used 18-20 days after planting and sacks are replaced after every 25 days.	Tamil Nadu
 Rice stem borer	To control stem borer in rice, 500 ml neem oil is mixed with 4 kg soil and some fresh cow dung. It is dried in shade for two days. There after it is dissolved in 50 litres water. About 200 g soap is dissolved in it and sprayed on the crop.	Orissa
	Broadcasting of peels of citrus fruits particularly of Rabab, Tenga (*Citrus grandis*) on standing water at mid tillering stage minimizes the damage of rice stem borers. Peels of citrus fruits repel various pests.	Andhra Pradesh
	Putting of bamboo sticks and or branches for birds sitting in the field in nursery and at tillering stage help pest reduction as bird catch the larvae and eat it away.	Uttar Pradesh and Orissa
	Release of *Trichogramma japonicum* @ 50,000 - 100,000 eggs/ha is very effective technique of stem borer management. These eggs are available in the form of *Tricho card*. Each *Tricho card* contains 20,000 eggs of parasite *Trichogramma*	

	japonicum. Tricho card can be further divided in 10 small pieces which can be easily tagged underside of leaf with the help of all pin / locally available thorn during evening hours. As per severity of stem borer damage *Trichogramma japonicum* can be released 5 - 6 times at 10 days interval.	
Gundhi bug or Rice bug or Ear head bug *Leptocorisa acuta* Thunberg.  Gundhi bug adult	Gundhi bug is a highly damaging insect of paddy crop that reportedly result in 80% yield loss. The insect sucks milk of the tender paddy grains at the initial stage. The grain formation is stopped and the paddy seeds remain hollow. Economic threshold level (ETL) of this pest is 1 bug / hill or 5 bugs/ m^2 in field condition.	
	Flowers of wild cactus (*Cycas circinalis*) commonly known as *Jangali madan mast ka phul* and *sannamboo* are tied in small paddy straw bundle and placed in rice fields to serve as repellent against ear head bugs[23]. The unpleasant odour emitted from the flower repels insect pests, especially ear head bug[24].	Nilgiri areas of Tamil Nadu
	Dusting of 5 kg rice bran with one litre of kerosene oil in the standing water of paddy field is reported to be effective to control ear head bug. The insect falling down may get drowned[5].	Uttar Pradesh
	The roots of *Achyranthus aspera* and bark of *Acacia lavcopholia* are dried well and powdered. It is mixed in water and sprayed to control ear head bug.	Uttar Pradesh
	A solution made of extract of 1kg garlic (*Allium sativum*), 200g tobacco leaves and 200g washing	Bareilly and Badaun districts of Uttar Pradesh

 Gundhi bug adult	powder dissolved in 200 litres of water is sprayed on the affected crop of paddy. One spray can control the insect by 80% [11].	
	Dusting of wood ash @ 15-20kg / acre control the bug. Ash contains boron, which is considered to have insecticidal properties.	Uttar Pradesh
	One meter high stand is erected in the paddy fields. During night one earthen lamp is placed on the stand after covering it with a funnel shaped iron sieve. Bugs are attracted to the light and get collected around the stand. Early in the morning farmers collect them and burry them under soil[17].	Eastern part of Uttar Pradesh
	Burning of straws of *Paspalum scrobiculatum* commonly known as *kodo, koda, koda dhan, kodra, arikalu, molvaagu, haraka, kodus* etc. and *Echinochloa frumentacea* commonly known as *sawan, sawa, shamula, samu, samul, bonta, somai, kudraivalipillu* etc., are burnt near paddy fields affected by gundhi bug. Insects get attracted and die. Some left alive are gathered physically and killed[17].	Eastern part of Uttar Pradesh
	Farmers control the *gundhi* bugs by generating smoke in the fields by burning herbs windward. Farmers also draw ropes saturated with resin and kerosene over the fields. Some scented aquatic plants like *Ceratophyllum demersum* Linn. *C. submersum* Linn.; *Lycopodium corinatum* Desb; *Limnophila* spp. and *Hydrilla verticilata* are also found useful to trap the gundhi bugs[25].	Orissa

 Gundhi bug adult	Farmers use bait of rotten crab mixed with furadan granule and packed in a fresh cloth to control *gundhi* bug. This bait is placed in different corners of the rice field. The *gundhi* bugs are attracted by the odour of rotten crabs, suck its juice and killed[5].	Gobindapur village in 24 Parganas of West Bengal
	Gundhi bug can be controlled by hanging a snail fish in a cloth attached to a stick erected in the field at about 25 places per acre. Due to rotten smell of snail, bugs leave the field.	Orissa
	Farmers put dead frogs or crabs in a bamboo stick equivalent to the length of the paddy, at milking stage to control bug infestation in paddy. Each dead frog/crab by and large can attract around 20 bugs. Attraction of bugs to the carcasses of frogs/crabs might be due to the smell of the carcass.	Khonoma village in Kohima district of Nagaland and tribal part of Orissa
Swarming caterpillar *Spodoptera mauritia* Boisduval.  Swarming caterpillar adult	The swarming caterpillars in paddy is controlled by broadcasting boiled rice mixed with hen blood after making in the form of pellets on the bunds and in the fields. Smell of blood and rice attract birds to the fields and they pick up the caterpillars[11].	Bihar
	Farmers spread the leaves of *Calotropis* spp in the standing crop of rice, which is infested by *Katara* (worm type of cutting insect larvae). Insects gather on the broadcasted leaves of *Calotropis* and next day all the leaves along with insect larvae are collected and destroyed and replaced by fresh leaves[26].	Mehsana area of Gujarat

Termite *Microtermes obesi* Holm.	Farmers know that standing water in fields helps in controlling termites. In paddy fields if there is no standing water, the termite attack is prevalent and drying up of paddy crop is the visible symptom. If cow dung manure is used extensively and there is water shortage, the termite infestation is common. Flooding helps in reducing the population of termites by disrupting their life cycle.	East Singhbhum, Jharkhand
	Termite can be controlled by seed treatment with chlorpyriphos 20EC @ 3.75 l /100 kg seed along with 10% solution of gum arabica just before sowing[10].	Orissa
	White ants/termites are very common in the rice field. White ants/termites are controlled by using a desi plough made of *neem* (*Azacirachta indica*) wood. It repels the insects and found effective in protecting rice crop.	Tamar block in Ranchi district of Jharkhand
Birds **House sparrow** *Passer domesticus* Linn. **Myna** *Acridotheres tristis* Linn. **Pigeon** *Columba livia* Gmelin **Parakeet** *Psittacula krameri* Scopoli. 	Birds, often visit paddy fields at grain ripening stage and threshing floors where they feed on grains. Tying cotton threads across the field, at the seedling stage in paddy field, can control white crane. By adopting this, bird's entry is stopped[11].	Ezhumatoor, Kopipuram and Puramattom villages in Pathanamthitta district of Kerala
	The audiocassette roles tied across the field along the borders in the field crops produce a sound, which creates phobia, in birds and acts as repellent, thereby avoiding the bird damage.	Gendoli village in Bundi district of Rajasthan
	Paddy seeds are properly mixed with cow dung before sowing. The mixing of the seed with cow dung becomes doubly advantageous. First, it protects the seed from birds whenever the nursery is dry and secondly it acts as manure for the seed.	Jammu region of Jammu and Kashmir

	At the time of milky to grain-filling stage keeping one effigy (a structure resembling human being made of cloth and paddy straw) per acre at the centre of the paddy field, one or two feet above the crop canopy by using bamboo sticks, the bird menace can be reduced. As the effigy appears like human, the birds fly off.	Uttar Pradesh, Bihar and Andhra Pradesh
	Scaring away sparrows from rice field is possible by putting the dark coloured pseudostem of *Colocassia* (*Colocasia esculenta*) in the shape of snake head.	Orissa
	Reflecting coloured ribbons may be used for scaring the birds [10].	Orissa
	Farmers tie the polythene sheet to a long pole place at the center of paddy fields with a small stick of 1.00 - 1.25m in the sides randomly. The sound made by the sheet due to wind make the bird pests fly away from the field thereby protecting the crop from bird menace.	Srikakulam areas of Andhra Pradesh
Field rat *Bandicota bengalensis* Gray and Hardwicke 	To control rats in the paddy field, cement mixed food is used. When eaten by the rats, it leads to stomach swelling, which leads to death of rats. About 90% the rats are controlled by this method.	Khanapur village in Belgaium district of Karantaka.
	Farmers fill broad-mouthed earthen pots with slurry made of cow dung upon which a little rice is spread to attract rats. When the rats try to eat rice, they instantly fall into the cow dung mixed water and die.	Orissa and Uttar Pradesh
	Use of jaggery dipped cotton balls controls rats. Small cotton balls are dipped in jaggery water and placed wherever rats are seen. When rats come in contact with the balls, they swallow them. Swollen cotton balls inside the stomach lead to death of the animal.	Koipuram, Ezhumatoor and Puramatoom blocks of Kerala

	Baiting with zinc phosphide is most effective in field as well as in house. Poison bait can be prepared by adding zinc phosphide 2 parts + shredded food grains 96 parts+ edible oil 1 part and sugar 1 part. Fumigation of rat burrows with aluminium phosphide tablet @ 1 tab./ burrow is also suggested to kill the field rat[10].	Orissa
	Leaves of *bhara/kans* (*Saccharum spontaneum*) are collected and placed in the opening of rat burrows in a continuous row on all sides of the field. The serrated margin of bhara/ kans cause injury to the rats and thus prevents rats from damaging fields.	Manda village in Bareilly district of Uttar Pradesh
Other indigenous techniques for rice pest control	Farmers dipped roots of rice seedlings in powdered solution of groundnut cake and neem cake before transplanting. The solution is prepared by soaking the cakes in water overnight. The seedlings so treated establish easily and are less vulnerable to pest attack for some period[27].	Pondicherry
	Deodar (*Cedrus deodara*) branches along with matured leaves are evenly spread over the paddy field. When the weeds come in contact with the branches and leaves, the oil present in deodar branches and leave work as a weedicide and eliminate the weeds in paddy field.	Hilly areas of Jammu and Kashmir
	Placing branches of *Calotropis gigantia* at the water inlet of paddy fields have been reported to control insect pests because of the alkaloid present in latex act as a repellent[8].	Karnataka

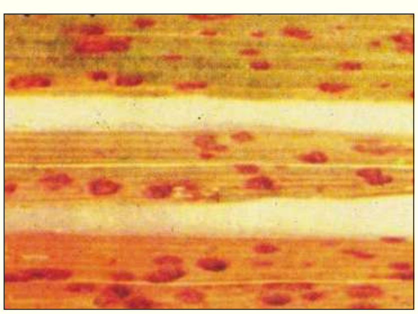
	Green and black aphid insects sit on *Calotropis procera* leaves. These leaves when placed in paddy fields attract aphids, and they sit on them instead of paddy leaves. Long branches with leaves of *Calotropis* should be placed at distance of 5m in paddy fields. Around 25-30 branches are required in one bigha (0.24 hectare). The branches dry up in ten days. These branches are changed thrice during which life span of aphids come to an end.	Anand and Kheda areas of Gujarat
	Before transplanting paddy seedlings are kept in small plots ($6m^2$) of standing water mixed with wood ash and pulverized neem seeds. Half kg of neem seed and 1 kg of wood ash is sufficient for mixing with water to accommodate 50 bundles of seedlings at a time for ½ to 1 hour. Treated seedlings produce healthy crops free from pests and diseases [28].	Perambalur area of Tamil Nadu
	The twigs of *Cleistanthus collinus* locally known as *karada* are used to control insect and pests in lowland rice cultivation. Tender branches of 1-1.5m length are planted erect all over the field at random after the crop is well established after transplanting or beushening operation, for direct-seed lowland rice, before the outbreak of any insect and pest incidents. The fresh leaves of this plant are applied on the standing water either in anticipation of the outbreak or at the initial stage of the incidence of insect-pest. This practice effectively controls insects and pests. The active principle has been identified as oduvin, a yellowish white crystalline glucoside.	Forest belt areas of West Bengal and Orissa

	Clipping off tip of rice seedlings before transplanting is practiced all over the state to ease transplantation; to facilitate uniform growth and to remove insect egg masses and other major insect pests present on the leaf-tips.	Assam
	Most of the paddy growers plant twigs of *kainth* (*Pyrus pashia*) in the periphery of the fields after paddy seedlings are sown. These keep away insect pests from damaging the paddy. It is seen that insect pests are reduced by 50%[11].	Rohru and Chirgaon villages in Shimla district of Himachal Pradesh
	Deodar oil is taken in a container. A stick is taken and one end of the stick is soaked in oil and the same is applied on the foliage of nursery plants. Before application, the water level in the nursery is raised. This causes caterpillars of paddy to crawl the foliage and the insect on coming in contact with the treated foliage is killed. After the application, the water is allowed to stand in the nursery for 1-2 hours[5].	Jammu region
	Neem leaves are mixed with equal quantity of *Cissus quadrongularis* leaves. The mixture is ground well and soaked in cow urine for one week and afterwards it is filtered. The filtrate is mixed with water at 1:9 ratio and it is sprayed twice at 15 days interval to control all the pests of paddy.	Jharkhand
	The fruits of *sausage tree* (*Kigelia pinnata*) are cut into pieces and buried in the soil of nursery area and main field, which considerably reduces the pest incidence in rice.	Perambur, Tamil Nadu

	Bhang (*Cannabis sativa*) plants are used for controlling threadworms in paddy nursery. *Bhang* plant is uprooted and kept in standing water of paddy nursery to control threadworm. If problem of threadworm is severe then crushed leaves are put in standing water to kill the worm [5].	Jammu and Kashmir
	Broadcasting of leaves and seed powder of custard apple (*Annona squamosa*) in field control the insect pests in paddy.	Jharkhand
	Cropping systems like rice - garlic-corn - tomato; rice-tobacco-rice and rice-garlic-maize-tomato are known to reduce the nematode population [29].	
	Green algae which occur in *kharif* paddy fields in stagnant water consumes oxygen from water of the rice field and produces carbon dioxide that results in yellowing and dwarfing of rice plants.It can be controlled by broadcasting 50-100 kg freshly plucked *karada* (*Cleisanthus collinus*) leaves in August. The toxic cations present in the *karada* leaves damage chlorophyll of green algae.	Panipila village in Nayagarh district of Orissa.
	The leaves and small twigs of *kochila* (*Strychnos nuxvomica*) are applied before land preparation, and subsequently incorporated into the soil to control weeds. Secretions of *kochila* twigs suppress 50-60% weeds inside the soil.	Anadpur in Keonjhar district of Orissa

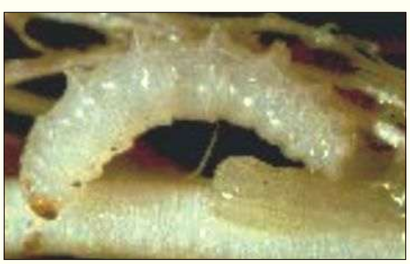
	Farmers dip the seeds of paddy in salted water before sowing. The floating seed are discarded along with the seeds of weeds. The water is drained out and the healthy seeds are kept in gunny bags for 24 hours for germination. The germinated seeds broadcasted in the field, where 5.0 - 7.5 cm standing water is maintained. Weed seeds fail to germinate under this condition. By using this practice, 70-80% weeds are reduced in the paddy crop.	Bulandshahar and Aligarh districts of Uttar Pradesh
	Small lamps and oil pans are kept in the paddy fields here and there during night-time. The nocturnal insects are attracted to the light and fall in the oil, resulting in killing of the insects. These lamps are locally available and this method is being practiced in all farming situations, which is cost effective and does not leave any residue.	Andhra Pradesh

Low cost disease management techniques in rice

 Rice blast	Farmwomen spray fresh cow dung extract to control bacterial leaf blight of paddy and it has been found very efficient. Economic threshold level (ETL) of this disease is 2-5% damaged leaf area in the field.	Kalahandi area of Orissa
	Farmers incorporate neem leaves into soil and puddle in the paddy nursery site, allow them to decompose for two weeks. After this treatment paddy seeds are sown and it is believed that seeds become tolerant to pest and disease[30 and 31].	Pondicherry
 Brown spot	Broadcasting of 20 kg salt / ha has been reported for the control of Khaira and blight diseases in paddy[32].	Gonda district of Uttar Pradesh

 Sheath blight	About 5 kg of *Lantana camara* leaves are soaked in 5 liters of water and 10 litres of cow urine for 3-4 days in a mud pot. It is then filtered and diluted in 80 litres of water. This solution is used for spraying and found highly effective in controlling fungal / bacterial disease[33]. Economic threshold level (ETL) of bacterial leaf blight is 2-5% damaged leaf area in the field.	Madurai area of Tamil Nadu
 Sheath rot	The roots of *Achyranthus aspera* commonly known as apamarg latjira, prickly chaff flower and bark of *Acacia lavcopholia* are dried well and powdered. They are mixed in water and sprayed to control leaf spot[5]. 2-5% affected tillers in the field is economic threshold level (ETL) of this disease.	Tiruchirapalli, Tamil Nadu
 Bacterial leaf blight affected plant	Leaves of wild tulsi, basil (*Ocimum basilicum*) plant can be used for controlling the blast disease of rice. About 1 kg tulsi leaves is boiled in 2 litres water thoroughly. The solution @ 2ml / l of water is stained and sprayed on the affected rice crop twice at 15 days interval [11]. Economic threshold level (ETL) of this disease is 5% damaged leaf area in the field.	Nimapada in Puri district of Orissa
 Young affected plant from Bacterial leaf blight	About 1 kg bael (*Aegle marmelos*) leaf is crushed and immersed in 10 litres of lukewarm water for 2 hours. Then the leaves are taken out and the solution is sprayed over the rice crop once daily for the control of blast in rice.	Mendhasal village in Khordha district of Orissa

 Sheath blight, rotting symptom on leaf	*Khaira* disease has been reported in many villages of Uttar Pradesh. The disease is caused by mineral deficiency. About 2-3kg of lime and one bucket of cowdung are mixed in 200 litres of water. This solution is sprayed on the affected crop of paddy. One spray controls the disease by 50%.	Bareilly and Badaun districts of Uttar Pradesh
	Paddy seeds are soaked in 20% *pudina* (*Mentha sativa*) leaves extract for 24 hours before sowing. This controls red leaf spot disease in paddy. This increases germination rate and vigor of seedlings[34] .	Tamil Nadu
	Rabbing (burning of nursery area) is done to minimize weed population in nursery bed; get healthy seedlings for transplanting and to increase resistance in rice plants against various diseases and insect pests. It helps in easy uprooting of seedling, burns weed seeds, adds organic matter and ash to the soil, resulting in vigorous growth of the seedlings. Soil borne fungi are inactivated due to *rabbing*. As a result, weed intensity is reduced by 80%, soil-borne disease 95%, stem borer incidence is reduced by 92%. The material is burnt during late evening hours. The nursery is raised in the burnt land.	Dang district of Gujarat and Sindhidurg district of Maharahstra
 Blast disease	Cowdung slurry is prepared by mixing 1kg cowdung with 10 litre water. The slurry is mixed with crushed *karada (Xylia xylocarpa)* leaves. The solution is sprayed at weekly interval to control blast in paddy.	Nimapada in Puri district of Orissa

Traditional techniques of Storage pest management **Rice weevil adult** *Sitophilus oryzae* Linnaeus 	With up to 40% of crops being damaged after they have been harvested, tackling this issue effectively will significantly improve food availability and security, as well as playing a vital role in reducing poverty. Proper storage of harvested crops is a major way to over come this issue.Angoumois grain moth *Sitotroga cerealella* Olivier, Rice weevil *Sitophilus oryzae* L.,Rice moth *Corcyra cephalonica* Stainton, Rice flour beetle *Tribolium castaneum* Herbst. and Lesser grain borer *Rhizopertha dominica* Fabr are the common storage pest in rice. They can be controlled by treatment of jute storage bags with malathion 50EC @ 5ml / 20 litre of water [10].	Orissa
	Dried neem leaves have been used effectively in paddy storage. About 200g of neem leaves are mixed with every 50kg of grains with a few of more tender branches of neem. Pests of any kind do not affect the grains stored with neem leaves for 2-3 months[35].	Kalahandi area of Orissa
 Rice weevil larva	To control moth and weevil infestation in paddy for every 50 kg of grain storage 200g crystals of common salt is placed. In a bag of 100kg paddy, 200g crystals of common salt are added after filling 50 kg and remaining 50 kg of grain is filled[11].	Orissa
	Rice grain is protected from storage pests by using common salt. By absorbing water, salt desiccates the insects to death. During filling the gunny bag with rice, 5-6 crystals of salt are put intermittently after each 10kg.	Danapur village in Bulandshahar district of Uttar Pradesh,

	Red pepper (*Capsicum* spp) or leaves of neyveli (*Ipomoea carnea fistulosa*) or dried chopped leaves of wild tobacco (*Chobelia nicotianafolia*) or neem leaves or coal fly ash or dried leaves of lantana @1kg /100 kg grain can be used for the protection of paddy from storage pests.	Sathyamangalam village in Puddukottai district of Tamil Nadu
	Mixing of garlic bulbs @1bulb/5kg rice is found effective for the reduction of storage pest damage in rice.	Bareilly, Shahjahanpur and Badaun districts of Uttar Pradesh
	Mixing of ash acts as a repellent (feeding deterrent). Ash contains boron, which is considered to have insecticidal properties.	Orissa
	Polishing process had drastic lethal and sub-lethal effects on the biology of *Sitophilus oryzae*. There is a delay in development of weevils in white rice compared with brown rice.	
Rice kept in Doli	Acetone extract of *bathua* (*Chenopodium album*) was found to have maximum repellency of 72.25% against rice moth *Corcyra cephalonica* during storage. *Chenopodium* extract is rich in terpenoids, which is suggested as the chemical factor for its repellent potential.	
 Rice kept in Akha over old vehicle tyre	Mixing *curry leaves* (*Murraya koenigii*) with grain produces unbearable odour and was found to minimize incidence of weevil and grain moth. Farmwomen use Jari akha for rice storage with the use of a mixture of *begonia* (*Vitex negundo*), *pudina* (*Mentha sativa*), *bhusunga patra, curry leaves* (*Murraya koenigii*), garlic and turmeric (*Curcuma aromatica*) powder.	Simor village in Khorda district of Orissa
	Covering grains with a layer of dried paddy husk of 5.0 – 7.5 cm minimizes incidence of weevil and grain moth.	Orissa

	Farmwomen store rice seeds in containers along with several layers of paddy straw and are found ideal for maintaining viability and less damage by storage pest[36].	Orissa
	Storing of rice in earthen pits for six months improved the quality of rice and saved from damage by store grain pests.	Orissa
	Storing of grains in *Puduka* is safe for seeds, which keeps it viable and free from store grain pests [37].	Nawapada district of Orissa
Dhan gada	Keeping of 10-15 red chilli fruits in one quintal rice bag prevent storage pests. The pungent odour of red chillies acts as a repellent [8].	Warrangal, Andhra Pradesh
	Mixing neem and eucalyptus leaves to food grains controls storage pests. Neem and eucalyptus act as insect repellents and antifeedants and oviposition deterrants[8].	Orissa and Uttar Pradesh
Selected earhead	Leaves of *arni* (*Clerodendron phlomides*) are used for preserving grains. Leaves are crushed to prepare its extract, which is bitter in taste. Five hundred ml of the extract is mixed with 40kg of grains. The grains are dried and filled in big earthen pots[38].	Hilly areas of Gujarat
 Farmwomen keeping selected earhead in polyethylene bag	Leaves of *notchi, neem and pongam (Pongamia pinnata*) are collected and put into the storage box or room along with the grains to protect them from pests and disease. The leaves can be changed regularly for better results. In another method, dried leaves of *notchi, neem and pongam* are put in a mud pot along with some dried chillies. These are then burnt and kept inside the storage room. The process is repeated every week depending upon the intensity of pest attack [39].	Uttar Pradesh

	In rural areas women spread a layer of leaves of walnut (*Juglans regia*) over grain stored in gunny bags. Likewise shade dried leaves of sweet flag are powdered and put over grain stored in gunny bags to protect it from damage due to stored grains pests. Walnut leaves are astringents and the aqueous extracts have bactericidal action while, mature leaves contain 9-11 % tannin, which are antifeeding agents[40] .	Himachal Pradesh
 Traditional storage in akha over stone and wood base	*Neem, behera* (*Texmiinalia bellirica*) walnut (*Juglans regia*) and mint leaves possess certain antimicrobial or pesticidal properties which help in grain storage. The turmeric powder also appears to perform a similar function. Grain is mixed with leaves of *neem*/ walnut/ behera/ mint (pudina) and then stored in bins [40] .	Himachal Pradesh
	Farmers mix the dried and powdered leaves of *banyan* tree (*Ficus benghalensis*) with the harvested grains for keeping it safe from pests. Sometimes *neem (Azadirachta indica*) leaves are also mixed with it [41]	Himachal Pradesh
	Farmers put common salt under the vessel containing the grains. Insects cannot move on the salt, thus their entry into the vessel is prevented. This method is cost effective.	Pondicherry

 Traditional storage in akha	The practice of storing the grains for seed purpose for next season with mixing of *neem* leaves or ash is generally followed to minimize the attack of stored grain pests on the seed. This practice is easy as no additional cost is involved. The scientific reason for this practice is that *neem* leaves have insecticidal, anti-feedant and repellent activity. The crystalline property of the ash, helps to create mechanical wound to the body wall of the insect as a result the dehydration takes place and the insect dies [32].	Uttar Pradesh
	Bark of *Paranus javanica* contains toxic principles used against stored rice pests.	Himachal Pradesh
	The leaves of wild mint (*Mentha species*) are dried and crushed into powder mixed with grain in the ratio of 1:100. This prevents the attack of stored grain insect pests.	Mandi district of Himachal Pradesh
	Rice, if kept along with *haldi, turmeric* (*Curcuma aromatica*) can be protected from moisture and insects .Farmers mix turmeric powder with rice for the purpose of reducing the insect damage. Turmeric powder acts as repellent for insect pests in rice.	Kapurthala district of Punjab and Uttar Pradesh
	A line of turmeric powder or mustard oil is drawn around the food grains/ storage bin or in the path of ants or other insect pests of stored grains. This practice repels the insects from stored grains.	Uttar Pradesh
Storage in separate room	Rice grain is protected from storage pests by using common salt. By absorbing water, salt desiccates the insects to death. During filling the gunny bag with rice, 5-6 crystals of salt are put intermittently after each 10kg. Treatment cost is Rs. 2 /100 kg of rice.	Bulandshahar district of Uttar Pradesh

 Mettalic drum used for storage	Rice is protected from storage pests by keeping a layer of match boxes before storage of rice in earthen containers and finally an other layer on the top of rice in storage container followed by sealing of the mouth of container.	Basti district of Uttar Pradesh
	Farmwomen mix turmeric powder with rice for the purpose of reducing the insect damage in rice. Turmeric powder acts as repellent for insect pests.	Kapurthala district of Punjab
	Fumigation of earthen storage containers by burning of neem leaves is a common practice to control storage pest in rice in the part of Aligarh district.	Uttar Pradesh
	About 5kg of crystal salt, 20 -30 red chilies and 5 -10 handfuls of dried neem leaves are thoroughly mixed with rice before bagging. Later the bags are also covered with dried neem branches. It minimizes the insect attack and improves the keeping quality of seed.	Orissa
 Kerosene oil drum used for storage	Paddy seed is stored in structures made of paddy straw rope, locally called *puduga* by mixing the grain with *neem or begonia* leaf @ 50g / 30kg seed for 1 year, thereby avoiding the damage to the seed from insects or diseases. Each structure can accommodate about 1 quintal seed.	Nawapada district of Orissa
	To avoid storage pest incidence in rice, the grains are mixed with leaves of *karkkurachi (Bassia latifolia*) and *aalimaram (Annona reticulate*) and then stored. This technology is very effective against storage pests.	Dindigul district of Tamil Nadu
	Seed coat of *mahua* (*Madhuca latifolia*) is used for storage of paddy seeds. 10-15 seed coat of *mahua* is mixed thoroughly in one kg of paddy seed before it is stored for seed purpose.	Bihar

 Rice storage in big tin container	The *ragi, finger millet* (*Eleusine coracana*) cobs have some odour and produce some chemical, which helps in keeping away insect pests from stored grains. The mature cobs of *ragi* are put inside the grain stores along with rice during storage. It is used @ 500g cobs /100kg grains kept in *pucca* houses having partition walls. About 85% grains are saved by this practice.	Aurangabad district of Bihar
	Farmwomen keep rice after drying and mixing of garlic for storage in Jari akha (Polyethylene bag) to save the rice from the attack of storage pests.	Padasahi and Kantamalim village in Khorda district of Orissa
	Paddy seeds are stored in box made of wood. Seeds of paddy are stored in *mora (pura)*. *Mora* is made of paddy straw and tied tightly with ropes made of paddy straw. Seeds can be stored for 5-10 years in *mora* where even rats cannot enter[42].	Orissa

Conclusion

Food is physiological need and eating food is a pleasure, therefore there is a demand for quality food which is safe for consumption. Crop pests are expected to be one of the major challenges to food security in the changing climate scenario. India can prevent crop losses amounting to Rs. 1, 50, 000 crore per year through judicious pest management and feed about 20% of its population with crop saved. Rapid usage of pesticides, often with insufficient technical advice and education has brought in its wake many environmental problems inimical to the interest of society. Besides excessive bio accumulation and build up in the biosystem and adverse effect on agricultural commodities has significant influence in the area of international trade. Keeping pace with the time, when environmental / health issues are agitating global minds, it should be our endeavour to phase-out chemical pesticides with safer techniques of botanical/ biological origins. Women play a crucial role in agriculture through out the world, producing, providing the food we eat. Despite their contribution to global food security they are frequently bypassed in various development strategies. Therefore, improving their economic status has now been recognized as an important component for any developmental activity. First women President of India Pratibha Patil in her very first official programme at National Academy of Agricultural Sciences, New Delhi advocated that our agriculture policy and programme should not be only pro poor and pro environment but it must be pro women. The technology of transfer for man alone being not sufficient for sustainable development therefore, it is the urgent need to focus cost effective and eco-friendly transfer of technology among women, so that farm women no longer have to share their hard earned produce with unwanted insect pests and diseases. Reorientation of agricultural research is considered imminent and is now a global priority in the context of plant protection and climate change. It is hoped that this documentation of low cost gender friendly techniques of pest management would serve as a reference guide for the agricultural field functionaries, particularly to the small holding producers where women play a significant role.

Scale and methodology for rice insect pest's observation[43]

Name of the Pest	Scale
Stem borer and Gallmidge	Select randomly 5 (five) places in a field and examine 5 hills in each location/count the number of tillers and dead heart/white ear head. Galls to be examined and percentage to be worked out and to be put up in a scoring 0-9 scale according to severity.
Score	0=nil; 1= less than 1%; 3= 2 - 4% ; 5= 5 -10%; 7 =11-20%; 9 = more than 20%
Stem borer moth	Number of moths to be disturbed by 10 steps inside the field and to be put in 0-9 scale.
Score	0 = nil; 1 = 1-2/hill; 3 = 3 - 5; 5 = 6-10; 7 = 11 - 20; 9 = more than 20.
Stem borer eggs	Stem borer eggs mass is to be recorded/clump while examining 25 clumps for stem borer dead heart/white ear head and is to be put in 0-9 scale.
Score	0 = none; 1 = less than 1; 5 = 1 - 2; 9 = more than 2.
Green leaf hopper	In all 10 clumps (5 clumps + 5 clumps at two places) diagonally across are to be examine and number of adults/hill can be recorded and put in 0-9 scale.
Score	0=nil; 1 = 5; 3 = 6-15; 5 = 16 - 30; 7 = 31 - 50; 9 = more than 50.
Brown plant hopper	By single knock to 20 hills at (10 + 10 clump at two places) diagonally across to be examined and counted/hill basis and put in 0 -9 scale.
Score	0 = nil; 1 = less than 1/hill; 3 = 2-4 / hill; 5 = 5-10/hill; 7 = 11-50/hill; 9 = more than 50/hill.
White backed plant hopper	Same as mentioned in brown plant hopper
Score	Same as mentioned in brown plant hopper
Leaf roller	25 hills are to be examined and damaged leaf hill should be counted and to be put in 0-9 scale.
Score	0 = no damage; 1 = less than 1 damaged leaf/hill; 3 = 1-2; 5 = 2-3; 7 = 4 -10; 9 = more than 10 damaged leaf/hill.
Leaf folder	Same as mentioned in leaf roller
Score	Same as mentioned in leaf roller
Rice hispa	Same as mentioned in leaf roller
Score	Same as mentioned in leaf roller

Scale and methodology for rice insect pest's observation[43]

Name of the Pest	Scale
Case worm	Same as mentioned in leaf roller
Score	Same as mentioned in leaf roller
Whorl maggot	25 clumps /hill are to be examined as in case of stem borer dead heart/white ear head and number of damaged leaves should be recorded according to 0-9 scale.
Score	0 = no damage; 1 = 2 damaged leaves; 3 = 5 damaged leaves; 5 = ½ of the total leaves damaged; 7 = ¾ of total leaves damaged; 9 = more than ¾ of total leaves damaged.
Army worm	By visual observation from the fields, the estimate should be given as per score.
Score	0= none;1=light;5=moderate;9=severe.
Climbing cutworm	Same as mentioned in army worm.
Score	0= none;1=light;5=moderate;9=severe.
Rice bug	Earheads of 10 hills are to be examined and number of bugs is to be counted as/hill basis.
Score	0 = none; 1 = 1 /hill; 3= 2-5 /hill; 5= 6-10/hill; 7 = 11-20 /hill; 9 = more than 20 number of bugs/hill.
Mealy bug	By visual observation from patches caused by attack of mealy bug and should be put in score.
Score	0 = no; 1 = 2% of area damaged; 3 = 2-4% of area damaged; 5=5% of area damaged; 7= 6-10% of area damaged; 9 = more than 10% of area damaged.
Thrips	By visual estimation, percent of damage.
Score	0 = none; 1= 5% light; 5 = 10-15% moderate; 9 = more than 50% severe.
Root weevil	Uproot the clump with soil and splash in water. Count the number of white grubs/hill by examining 3 numbers of clumps. The average number of grub/plant/hill should be recorded and reproduced in scale 0-9.
Score	0 = none; 1=1/hill; 3 = 2-5/hill; 5 =6-10/hill; 7=15-20/hill;9= more than 20/ hill.
Grass hopper	Same as mentioned in stem borer moth.
Score	Same as mentioned in stem borer moth.

Scale and methodology for rice disease observation[43]

Name of the disease	Scale
Leaf blast *Pyricularia oryzae* Symptoms:Typical leaf lesions are spindle shaped and often develop grayish centers and coalesce on susceptible plants	0 = no lesions 1 = small brown specks 2 = larger brown specks 3 = small, roundish to slightly elongated, necrotic grey spots, about 1-2 mm in diameter with brown margin 4 = typical blast lesion elliptical 1-2cm long, usually confined to the area of the two main veins, less than 2% of leaf area infected with typical blast lesions 5 = less than 10% of leaf area infected with typical blast lesions 6 = average of about 25% of leaf area infected with typical blast lesions 7 = average of about 50% of leaf area infected with typical blast lesions 8 = average of about 75% of leaf area infected with typical blast lesions 9 = about 100% of leaf are infected.
Neck and node blast *Pyricularia oryzae* Symptoms: Panicle neck or branch with dark necrotic lesions and frequently broken at the point of infection. Panicles in such cases are either partially filled or have no grain and are grayish if infected early.	0 = no incidence 1 = less than 1% panicles infected 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = 51-100%
Brown spot *Cochliobolus miyabeanus* *Helminthosporium oryzae* Symptoms: Typical leaf spots are small, oval or circular and dark brown large lesions are usually the same colour on the edges but have a pale, usually grayish center. Most spots have slight yellow hollow around the outer edge. Two digits are used to measure; on the basis of lesion type and on the basis of severity/leaf area affected.	<u>1. On the basis of lesion type</u> 1 = pinhead spots type lesion 5 = typical brown spots type lesion sometime with grey centre 9 = large brown spots with grey centre <u>2. On the basis of severity/leaf area affected</u> 1 = less than 1% 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = more than 51%

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Name of the disease	Scale
Narrow brown leaf spot *Cercospora oryzae* Symptoms : Narrow, reddish brown spots parallel to the leaf veins	0 = no incidence 1 = less than 1% leaf area affected 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = more than 50%
Bacterial leaf streak *Xanthomonas translucens* F. sp. *oryzae*	0 = no incidence 1 = less than 1% leaf area affected 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = more than 50%
Leaf smut *Entyloma oryzae*	0 = no incidence 1 = less than 1% leaf area affected 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = more than 50%
Leaf scale *Rhynchosporium oryzae* Symptoms: The lesion occurs mostly near the tips, but sometimes starts at the margin of the blade and develops into large ellipsoid areas encircled by dark brown, narrow banes accompanied by light brown hollow.	Leaf area affected 0 = no incidence 1 = less than 1% (apical lesions) 3 = 1-5% (apical lesions) 5 = 6-25% (apical and some marginal lesions) 7 = 26-50% (apical and some marginal lesions) 9 = more than 50% (apical and marginal lesions)
Bacterial blight *Xanthomonas oryzae* Symptoms: Lesions usually start near the leaf tips and / or leaf margins and extend down the outer edge(s). Young lesions are pale-green to grey-green, later turning yellow to grey (dead) with time. Very susceptible varieties may have lesions extending the entire length even into the leaf sheath.	0 = no incidence 1 = less than 1% hills affected 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = more than 50%

Scale and methodology for rice disease observation[43]

Name of the disease	Scale
Leaf blight	0 = no incidence 1 = less than 1% leaf blade area showing necrotic symtoms 3 = 1-5% 5 = 6-25% 7 = 26-50% 9 = more than 50%
Rice tungro virus Symptoms: Leaves yellow to orange yellow, stunting and slightly reduced tillering	0 = no incidence 1 = less than 1% 2 = about 5% 3 = about 10% 4 = about 20% 5 = about 30% hills infested 6 = about 40% 7 = about 60% 8 = about 80% 9 = about 100% hills infested.
Grassy stunt virus Symptoms: Pale green, erect leaves, sometimes with blotches, excessive tillering and stunting.	Same as mentioned in rice tungro virus
Sheath blight *Thanatephorus cucumeris* Symptoms: Greyish green lesions may enlarge and coalesce with other lesions mostly on lower leaf sheath but occasionally on the leaves.	0 = no incidence 1 = lesions limited to lower ¼ of leaf sheath 3 = lesions present on lower ½ of leaf sheath. 5 = lesions present on more than ½ of leaf sheath, slight infection on lower (3rd or 4th) leaves. 7 = lesions present on more than ¾ of leaf sheath. Severe infection on lower leaves and slight infection on upper leaves (flag and 2nd leaf) 9 = lesions reaching top of tillers; severe infection on all leaves.
Sheath rot *Acrocylindrium oryzae* Symptoms : Oblong or irregular brown to grey lesions on the leaf sheath near panicle; sometimes coaleascing to kill emerging young panicle.	0 = no incidence 1 = less than 1% tillers affected 3 = 1 to 5% 5 = 6 to 25% 7 = 26 to 50% 9 = 51 to 100%

Scale and methodology for rice disease observation[43]

Name of the disease	Scale
False smut *Ustilaginoides virens* Symptoms : infected grains are transformed into yellow greenish or greenish black spore balls of a velvety appearance	0 = no incidence 1 = less than 1% glumes discoloured/panicle 3 = 1-5% 5 = 6-25% 7 = 26 -50% 9 = more than 50%
Zinc deficiency Symptoms: Leaves with brown spotting non-pathological. At earlier stages the mid-rib turns white. Plants stunted.	0 = no spot 1 = spots scantly and light. 5 = spots upto 50% of lamina 9 = entire leaf affected.